

Zavier Kamath

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Local Address

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Summary

- Xavier Kamath is an **aspiring astrophysicist** with an interest in **cosmology**, **dark matter**, and **structure formation**

Education

The Ohio State University May 2025 (Expected)

- Cumulative **GPA: 3.972**
- Major 1: **Astronomy and Astrophysics**
- Major 2: **Physics**
- Minor: **Spanish**

Relevant Coursework:

- **Mathematics:** Calculus 1-3, Linear Algebra, ODEs and PDEs
- **Physics:** Quantum Mechanics 1 & 2, Intermediate Mechanics 1 & 2, Statistical Mechanics, E&M 1, Experimental Physics Instrum. and Data Analysis Lab, Big Data Analytics Lab
- **Astronomy:** Astron. Data Science, Intro Cosmology, Stellar, Galactic, and Extragalactic Astrophysics, Methods of Astron. Observation and Data Analysis
- **Expected (Before Graduation):** Advanced Astronomy Data Analytics, Advanced Physics Lab, Computational Physics

Research Experience

Research Project Under Prof. Annika Peter and Prof. Ivan Esteban

(Summer 2024)

Remote Work

- Continuation of 2023 summer work. **Advanced statistical analysis** (Monte Carlo Estimates, Unbinned Likelihood) of how Warm Dark Matter (WDM) models on small scales match observational data from Milky Way satellite galaxies
- State-of-the-art use of statistics in an astronomy, physics, and cosmology context
- **Creation of well-factored and performant Python code** to carry out advanced statistical analysis
- Contribution of professional writing of methods and conclusions to be included in an **in-progress publication**
- Result of project was a further constraint of viable WDM models
- Github page created with an organized presentation of all coding methods, scripts / utilities, and plots

Physics Summer Research Program (SRP)

(Summer 2023)

The Ohio State University, Columbus, OH

- Collaborated on a theory-focused project working under **Prof. Annika Peter** and **Dr. Ivan Esteban** aiming to compare **dark matter** model (WDM) predictions to observational data from Milky Way satellite galaxies
- Extensive usage of Python programming and analysis and creation of detailed plots
- Presentation of findings at SRP **poster session**

Organizations

Astronomical Society (Autumn 2023, The Ohio State University)

- Completion of professional development in **operating radio telescopes** at the **Greenbank Observatory**

Buckeye Gaming Collective (Autumn 2023 - Spring 2024, The Ohio State University)

- **Captain** of VALORANT club team

Skills

- **Python** (NumPy, SciPy, Matplotlib, Pandas, SciKit-Learn, Plotly, Vegas) for **data analytics**, **machine learning**, and **statistical modeling** in astrophysics
- Advanced knowledge of statistics, astronomy, physics, and mathematics concepts
- Basic **Linux terminal** operation knowledge
- **LaTeX** for scientific writing
- Practiced in reading and critically thinking about scientific papers
- Conversational in **Spanish**

Honors / Awards

2023 Physics Summer Undergraduate Research Scholarship

- Funded by Captain Forest R. Biard Undergraduate Research Scholarship Fund in Physics

Trustees Scholarship (Fall 2021 - Spring 2025)

- Merit-based scholarship awarded by The Ohio State University

Dean's List (Fall 2021 - Spring 2024)

- For all 6 semesters so far at The Ohio State University